

Network Science (Barabasi')

(networksciencebook.com)

Practice Problems

① Modelling a collaboration network.

A mid-sized company analyzes weekly collaboration between 6 units:

Sales (S), Marketing (M), Logistics (L), IT (I)
Finance (F), HR (H).

The collaboration relations (links) are:

→ (S,M), (S,L), (M,I), (L,I), (I,F), (F,H).

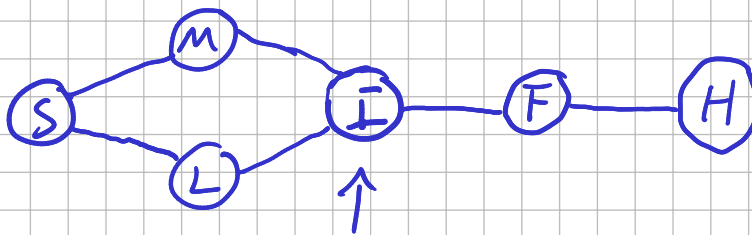
a) Define the node set (V) and edges (E)

$$G = (N/V, E)$$

$$V = \{S, M, L, I, F, H\}$$

$$E = \{ \text{ " " } \}$$

b)



* Bottleneck

c) Vocabulary: Adjacency matrix (Adjazenz)

$$A = \begin{array}{c|cccccc} & S & M & L & I & F & H \\ \hline S & 0 & 1 & 1 & 0 & 0 & 0 \\ M & 1 & 0 & 0 & 1 & 0 & 0 \\ L & 1 & 0 & 0 & 1 & 0 & 0 \\ I & 0 & 1 & 1 & 0 & 1 & 0 \\ F & 0 & 0 & 0 & 1 & 0 & 1 \\ H & 0 & 0 & 0 & 0 & 1 & 0 \end{array}$$

d) Determine n , m , degree of each node, avg. degree

$$n = 6 \quad (\# \text{ nodes})$$

$$m = 6 \quad (\# \text{ edges})$$

$$K_S = 2$$

$$K_M = 2$$

$$K_L = 2$$

$$K_I = 3$$

$$K_F = 2$$

$$K_H = 1$$

Avg. degree

$$\langle k \rangle = \frac{2m}{n}$$

* Important!

- In an undirected graph, each edge connects 2 nodes

$$\frac{2m}{n} = \frac{2 \cdot 6}{6} = \underline{2}$$

$$\langle k \rangle = \frac{1}{6} (2+2+2+3+2+1) \quad \text{Alternative:}$$

$$= \frac{1}{6} (12) = \underline{2}$$

$$\langle k \rangle = \frac{1}{n} \sum_{i=1}^n K_i$$

Laplacian:

$$L = D - A$$

D =

	S	M	L	I	F	H
S	2	0	0	0	0	0
M	0	2	0	0	0	0
L	0	0	2	0	0	0
I	0	0	0	3	0	0
F	0	0	0	0	2	0
H	0	0	0	0	0	1

A =

	S	M	L	I	F	H
S	0	1	1	0	0	0
M	1	0	0	1	0	0
L	1	0	0	1	0	0
I	0	1	1	0	1	0
F	0	0	0	1	0	1
H	0	0	0	0	1	0

L =

	S	M	L	I	F	H
S	2	-1	-1	0	0	0
M	-1	2	0	-1	0	0
L	-1	0	2	-1	0	0
I	0	-1	-1	3	-1	0
F	0	0	0	-1	2	-1
H	0	0	0	0	-1	1

③ Shortest paths in a product co-purchase network.

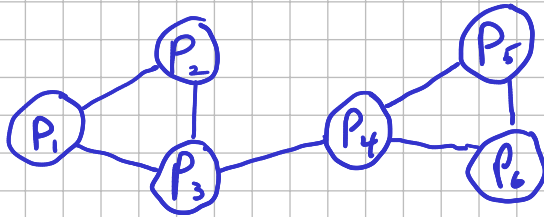
An online retailer observes the following co-purchase network of 6 products:

$(P_1, P_2), (P_1, P_3), (P_2, P_3), (P_3, P_4), (P_4, P_5), (P_4, P_6), (P_5, P_6)$.

a) Draw the graph.

$$V = N = \{P_1, P_2, P_3, P_4, P_5, P_6\}$$

$$E = \{ \text{ " " } \}$$



b) Compute the shortest-path distances between P_1 and the other nodes

$$d_{12} = 1$$

$$d_{13} = 1$$

$$d_{14} = d_{13} + d_{34} = 2$$

$$d_{15} = d_{13} + d_{34} + d_{45} = 3$$

c) " " from P_4 to the other nodes.

$$d_{41} = 2$$

$$d_{42} = 2$$

$$d_{43} = 1$$

$$d_{45} = 1$$

$$d_{46} = 1$$

d) Determine the longest shortest-path distance (diameter)

$$d_{16} = d_{15} = d_{26} = d_{25} = \underline{\underline{3}}$$

e) Determine the APL.

$$APL = \frac{1}{N(N-1)} \cdot \sum_{i,j=1}^n d_{ij}$$